TypeScript Foundations: Essentials for JavaScript Developers

Understanding the Why of TypeScript

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Our GitHub Repo

https://github.com/FeynmanFan/pstypescriptcc

I Love TypeScript

I'm about 50/50 on Microsoft

I've been working with JavaScript since the very beginning

How JavaScript Developed

- Possibly the most successful programming language of all time
- But it developed in an ad hoc way
- Good, but also bad
- TypeScript is an effort to place a design ethos atop JavaScript
- Especially a type system – "Type"Script
- Allow developers to leverage the language tools they know

So, Is TypeScript Just C# on top of JavaScript?

No

Another gem from Anders Hejlsburg

TypeScript is enhancing JS, not supplanting it

"Pythonic"

TypeScript is JavaScript-ic

Preserving what design ethos there is

Does Typing Matter?

"Static" or "Strong" Typing…not really the same thing

Focus on "type coercion"

Simple Type Coercion in JavaScript

```
var x = 5;
x = “drum solo”;
console.log(x);
```

Type Coercion

Sometimes it's useful to coerce to a new type

Strict typing is *commitment*

I think that type coercion is often a code smell

Why else recommend const over var?

Type coercion happens more often due to an error in coding than by design with loose typing

[1,2,3]

Unit Testing and Typing

```
$employee.Birthday = “This is not a birthday and will break the code that tries to use it”;

employee.Birthday = “This is not a birthday and will break the code that tries to use it”;

employee.Birthday = “This is not a birthday and will break the code that tries to use it”;

employee.Birthday =”2024-06-01”;

employee.Birthday = DateTime.Parse(”2024-06-01”);

// Does TypeScript enforce strong typing?
```

TypeScript Type Enforcement

```
x = 5;
x = “drum solo”;

// TypeScript type enforcement happens at compilation time
// only, and only for TypeScript code.
```

Compilation vs. Transpilation

The JavaScript emitted from processing TypeScript is at the same level of abstraction as TypeScript.

Version Controlling TypeScript and JavaScript

Keeping Binaries in Version Control Stinks

Craft your .gitignore

Keep the junk out of version control

Some stuff changes every time we build – that's junk

What is the principle?

We version control only the purely non-deterministic elements of the system.

Deterministic

```
346 x 183
= 63318

// Today, tomorrow, now and forever
// 63318 is the deterministic outcome of 346 * 183
```

An Exception to the Rule

Partial migration

You can't just add *.js to gitignore

The price of the transition

Demo: How to Version Control TypeScript

Demo: Debugging TypeScript